ITCS 254: Introduction to Linux, Shell, and Commands

Lab 9 – User Environment

1. Create an alias of “ls -la” command as “lla” and test it.

```
root@ubuntu:~$ alias lla='ls -la'
root@ubuntu:~$ lla
total 32
drwx------  4 root root 4096 Oct  3 16:31 .
drwxr-xr-x 22 root root 4096 Oct  3 16:30 ..
-rw-r--r--  1 root root 3234 Oct  3 16:31 .bashrc
-rw-r--r--  1 root root  161 Apr 22  2024 .profile
drwx------  2 root root 4096 Oct  3 16:29 .ssh
drwxr-xr-x  7 root root 4096 Oct  5 09:57 .theia
-rw-r--r--  1 root root  109 Oct  3 16:31 .vimrc
-rw-r--r--  1 root root  165 Oct  3 16:31 .wget-hsts
lrwxrwxrwx  1 root root    1 Oct  3 16:30 filesystem -> /
root@ubuntu:~$ ls -la
total 32
drwx------  4 root root 4096 Oct  3 16:31 .
drwxr-xr-x 22 root root 4096 Oct  3 16:30 ..
-rw-r--r--  1 root root 3234 Oct  3 16:31 .bashrc
-rw-r--r--  1 root root  161 Apr 22  2024 .profile
drwx------  2 root root 4096 Oct  3 16:29 .ssh
drwxr-xr-x  7 root root 4096 Oct  5 09:57 .theia
-rw-r--r--  1 root root  109 Oct  3 16:31 .vimrc
-rw-r--r--  1 root root  165 Oct  3 16:31 .wget-hsts
lrwxrwxrwx  1 root root    1 Oct  3 16:30 filesystem -> /
root@ubuntu:~$ alias
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias lla='ls -la'
alias ls='ls --color=auto'
```

2. Create a user “bob”.

```
root@ubuntu:~$ useradd bob
root@ubuntu:~$ cat /etc/passwd|grep bob
bob:x:1001:1001::/home/bob:/bin/sh
```

3. Define the password of bob as “abc123”.

```
root@ubuntu:~$ groups bob
bob : bob
root@ubuntu:~$ passwd bob
New password:
Retype new password:
passwd: password updated successfully
root@ubuntu:~$
```

4. Check /etc/passwd and /etc/group files after creating user bob.

```
root@ubuntu:~$ cat /etc/passwd|grep bob
bob:x:1001:1001::/home/bob:/bin/sh
root@ubuntu:~$ cat /etc/group|grep bob
bob:x:1001:
root@ubuntu:~$ groups bob
bob : bob
```

5. Check which group bob belongs to

```
root@ubuntu:~$ cat /etc/group|grep bob
bob:x:1001:
```

6. Create two group named “dev1” and “dev2”.

```
root@ubuntu:~$ groupadd dev1
root@ubuntu:~$ groupadd dev2
```

7. Assign user bob to both group dev1 and dev2.

```
root@ubuntu:~$ usermod -a -G dev1 bob
root@ubuntu:~$ usermod -a -G dev2 bob
root@ubuntu:~$ groups bob
bob : bob dev1 dev2
root@ubuntu:~$ id bob
uid=1001(bob) gid=1001(bob) groups=1001(bob),1002(dev1),1003(dev2)
```

8. Check again which group bob belongs to and see some changes in /etc/passwd and /etc/group files

```
root@ubuntu:~$ cat /etc/passwd|grep bob
bob:x:1001:1001::/home/bob:/bin/sh
```

```
root@ubuntu:~$ cat /etc/group|grep bob
bob:x:1001:
dev1:x:1002:bob
dev2:x:1003:bob
```

9. Change bob to be in group dev1 only.

```
root@ubuntu:~$ usermod -G dev1 bob
root@ubuntu:~$ groups bob
bob : bob dev1
```

10. Switch to user bob.

```
root@ubuntu:~$ su bob
\[\033[01;32m\]\u@\h\[\033[00m\]:\[\033[01;34m\]\w\[\033[00m\]$
```

11. Switch to root.

12. Create an empty file named test.txt and check its permissions.

```
root@ubuntu:~$ touch test.txt
root@ubuntu:~$ chown bob test.txt
root@ubuntu:~$ ls -l
total 0
lrwxrwxrwx 1 root root 1 Oct  3 16:30 filesystem -> /
-rw-r--r-- 1 bob  root 0 Oct  5 10:24 test.txt
```

13. Change bob to be the owner and dev2 to be the group owner of test.txt

```
root@ubuntu:~$ chgrp dev2 test.txt
root@ubuntu:~$ ls -l
total 0
lrwxrwxrwx 1 root root 1 Oct  3 16:30 filesystem -> /
-rw-r--r-- 1 bob  dev2 0 Oct  5 10:24 test.txt
```

14. Change permission of test.txt to be `rwxrw-r--` by numeric mode and display output.
15. Change permission of test.txt to be `rw-rw-rw-` by symbolic mode and display output.